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Research on the construction and application of teacher-student interaction evaluation system for smart classroom in the post COVID-19

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ABSTRACT

Smart classroom (SC) is an important embodiment of the new teaching mode in online and offline integration in the COVID-19. At present, the research on the evaluation system of teacher-student interaction (TSI) in SC lacks evaluation system to guide the development of students' higher order thinking (HOT). This evaluation system of TSI was established through grounded theory, Delphi method and analytic hierarchy process. The evaluation system for TSI considered the support of intelligent technology for various indicators and the development of students' HOT. The evaluation system included 4 first level indicators, 16 s level indicators and 44 third level indicators. The evaluation system had been comprehensively evaluated the TSI activities, which was conducive to the development of students' HOT.
Data availability: Data used in this research are available upon request from the corresponding author.

1. Introduction

In the spring semester of 2020, in order to effectively respond to the impact of the COVID-19 and achieve non-stop classes, 180 million primary and secondary students and 10 million teachers implemented multi-level, full course, and regular online education (Zong et al., 2020). This was not only a test of China's education informatization construction, but also a major opportunity to promote the education reform in the information age (Xue et al., 2022). Online education has achieved positive results during the COVID-19: the integrated teaching space framework of "online and offline, on and off campus, home school integration" has been basically formed, the concept and quality of teachers and students' information education have been greatly improved, and the way of teaching and learning has changed significantly, and the integrated application of information technology and education and teaching has been effectively carried out (Huang & Tang, 2020). The new teaching model of online and offline integration needs to be actively explored to create and build smart classroom (SC) in the post COVID-19, and the SC needs to be constructed. SC integrates information technology, intelligent technology and thinking teaching design technology to ensure a rich and effective learning environment for teacher-student interaction (TSI). The TSI is the core element of the system from the perspective of the digital learning environment, which is the key for the environment to have the function of technology effectively acting on the development of thinking (Zhang & Tian, 2022). Therefore, the key point of building a high-quality and effective SC is TSI. Studying the TSI and exploring the evaluation system of TSI in the SC under the COVID-19 conditions have important theoretical value and practical significance for effectively developing students' higher order thinking (HOT) and even cultivating innovative talents.

The evaluation tool of TSI is an important research field of TSI. Since the 1960 s, different evaluation tools have been produced according to different goals. Typical evaluation tools for TSI include Flanders interactive analysis system (FIAS) (Flanders, 1963), teacher-student analysis method (Xu & Yang, 2019), Trends in international mathematics and science study video assessment framework (Marian & Jackson, 2017) and classroom assessment scoring system (Buell et al., 2017). After information technology was introduced into the classroom, the TSI evaluation tool containing technical elements came into being. A typical example is the information technology-based interaction analysis system (ITIAS) for teacher-student (Yuan, 2022). With the development of technology, SC has gradually become the focus of research. Based on FIAS and ITIAS and the technical characteristics of SC, researchers have designed an evaluation tool for TSI in SC, which include four first level

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indicators of teacher's language behavior, student's language behavior, silence, teacher's use technology and student's use technology, as well as 18–24 s level indicators (He et al., 2019; Zhang et al., 2016; Shi et al., 2019).

The existing evaluation tools for TSI have four problems. First, the evaluation dimension is relatively single, which is based on speech interaction. Second, the main goal is to evaluate students' knowledge, and pay less attention to the development of students' HOT. Third, the acquisition method of evaluation dimension is relatively simple, and the top-down deduction method is adopted based on FIAS. Fourth, each evaluation dimension lacks the support of intelligent technology. Compared with traditional classroom and simple multimedia classroom, TSI in SC is highly complex and variable, which reflects the characteristics of higher order interaction goals, multi-dimensional interaction content, diversified interaction methods and rich and intelligent interaction media. The existing evaluation tools can hardly meet the practical needs of TSI in SCs. This research adopts grounded theory, Delphi method and analytic hierarchy process (AHP). The research objective is to (1) construct the evaluation index system mainly for the nonverbal interaction between teachers and students from multiple dimensions, (2) achieve the core goal of developing students' HOT, (3) realize bottom-up method induction and extraction of evaluation dimensions, (4) fully reflect the supporting role of intelligent technology on each evaluation dimension. The evaluation system will guide teachers to build high-quality and effective SC and help students develop HOT.

2. Method

2.1. Selection of evaluation index system construction methods

This research chose the grounded theory method to construct the evaluation system of TSI in SC. The grounded theoretical method was a qualitative research method under the empirical research paradigm, which required researchers to go deep into the activity site to collect many first-hand materials. These collected raw materials were coded, compared, classified, and synthesized under the relevant context and social background, and finally the empirical data were constructed into their own substantive theories. This method was applicable to the theoretical construction when there was no existing theory to explain or reference the research topic (Wang et al., 2022). Since the evaluation system of TSI in SC was a new research field, and there was no mature theoretical framework for direct reference, this study chose a grounded theoretical method to obtain evaluation indicators from bottom to up.

The grounded theory method was further divided into classical grounded theory and procedural grounded theory, the former emphasized continuous comparative analysis without prior theoretical assumptions to make the theory emerge naturally, while the latter needed to be based on certain prior theoretical assumptions (Wu et al., 2016). Since researchers had read a lot of relevant literature on TSI and SC in advance, they had formed a certain theoretical hypothesis, and selected the programmed grounded theory method to extract the evaluation indicators of TSI in SC. This method required researchers to conduct in-depth research on the literature to form theoretical sensitivity and maintain insight into theoretical terms. This method analyzed the coding of research materials, which included open coding, spindle coding, selective coding, theoretical saturation test and other steps (Chen et al., 2018). The operation process included determining research questions, literature research, theoretical sampling, research interviews, coding process, continuous analysis, saturation test, generation theory, etc. Data collection and analysis was a cycle process until theoretical saturation was reached. Research process of programmed rooting theory method is shown in Fig. 1 (Zhang & Zhang, 2018).

2.2. Data collection

This research selects 45 typical SC teaching videos in the "National information technology and curriculum integration" contest as the research object. Two thirds (30) of these videos were randomly selected for theoretical analysis, and one third (15) of the videos were reserved for theoretical saturation test. Because the research aimed to explore an operational evaluation system for guiding the construction of SC from the perspective of the elements of TSI. The selected videos included mathematics, physics, chemistry, English, science, comprehensive practice, etc., which included the curriculum of primary school, junior high school, and senior high school (Table S1).

2.3. Ethical and moral explanation

Researchers promptly communicated with relevant school administrators and teachers after determining the research subjects. The researcher provided detailed information on the objective and theme of this study, which fully respected the wishes and requirements of the respondents. The researcher also promised to the respondents that the research materials only were used for academic research and were not used for other objectives. The final research results did not involve the specific institutions of the respondents. After the coding of the research materials was completed and before the formal article was formed, the main materials were resubmitted to the respondents for review and confirmation before being officially published.

2.4. Coding and analysis

To ensure the scientific and accuracy of coding analysis, NVivo11 qualitative analysis software from QSR company (Alfadil et al., 2020) was used to encode and analyze teaching videos one by one. Two researchers conducted coding and category conceptualization back-to-back. After completing the encoding of each teaching video, researchers immediately checked and proofread to reach a consensus before proceeding to the encoding process of the next teaching video. The theory obtained by this analysis method had strong explanatory power.

2.4.1. Open coding and axial coding

The purpose of open coding was to discover phenomena from raw

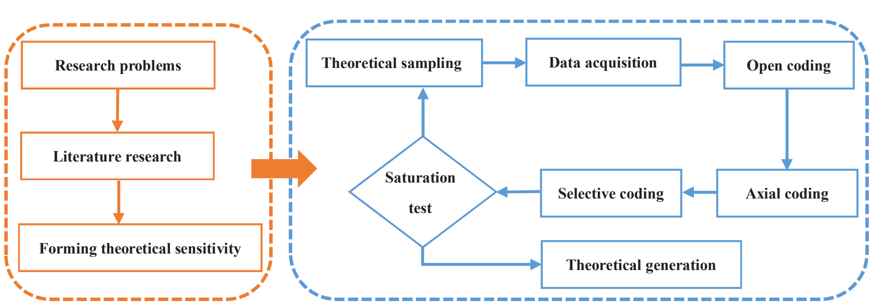

Fig. 1. Research process of programmed rooting theory method.

materials and extract initial concepts (Li & Wu, 2022). According to the operation process of grounded theory method, NVivo11 qualitative analysis software was used to mark teaching videos, encode fragments, create free nodes, modify nodes, compare and summarize nodes, and merge nodes with the same meaning. A total of 477 reference points and 213 initial concepts were extracted, and then these initial concepts were further categorized to form 41 initial categories. The open coding of TSI in SC is shown in Table S2.

Axial coding was a process to further deepen the many initial categories formed by open coding, which aimed to get more general or abstract categories. The main task was to discover and construct the relationship between categories including causality, time relationship, precedence relationship, situational relationship, similarity relationship, etc., so the initial concepts obtained in the previous stage were organically linked (Sawatsky et al., 2019; Balmer & Richards, 2017). This study summarized and improved the 41 initial categories obtained by open coding, which obtained 16 categories and the relationship between the initial categories and categories. Each category contained 2–3 initial categories.

2.4.2. *Selective coding and theoretical saturation test*

This research closely centered on the core theme of “evaluation system of TSI in SC”, combed the logical relationship between it and categories with “story line”, and further classified the 16 categories into four categories: operational interaction, behavioral interaction, cognitive interaction, and innovative interaction. Operational interaction, behavioral interaction, cognitive interaction, and innovative interaction included four categories respectively. Each category specifically included 2–3 initial categories, and the whole system included 41 initial categories. In this study, four categories, 16 categories, 41 initial categories and their relationships were used to build an evaluation system for TSI in SCs (Fig. 2). A total of 1/3 (15) teaching videos were reserved for theoretical saturation test. Except for the above initial categories and categories, no new initial categories and categories were found in these teaching videos. Therefore, the evaluation system of TSI in SC had reached theoretical saturation (Zhan et al., 2021a).

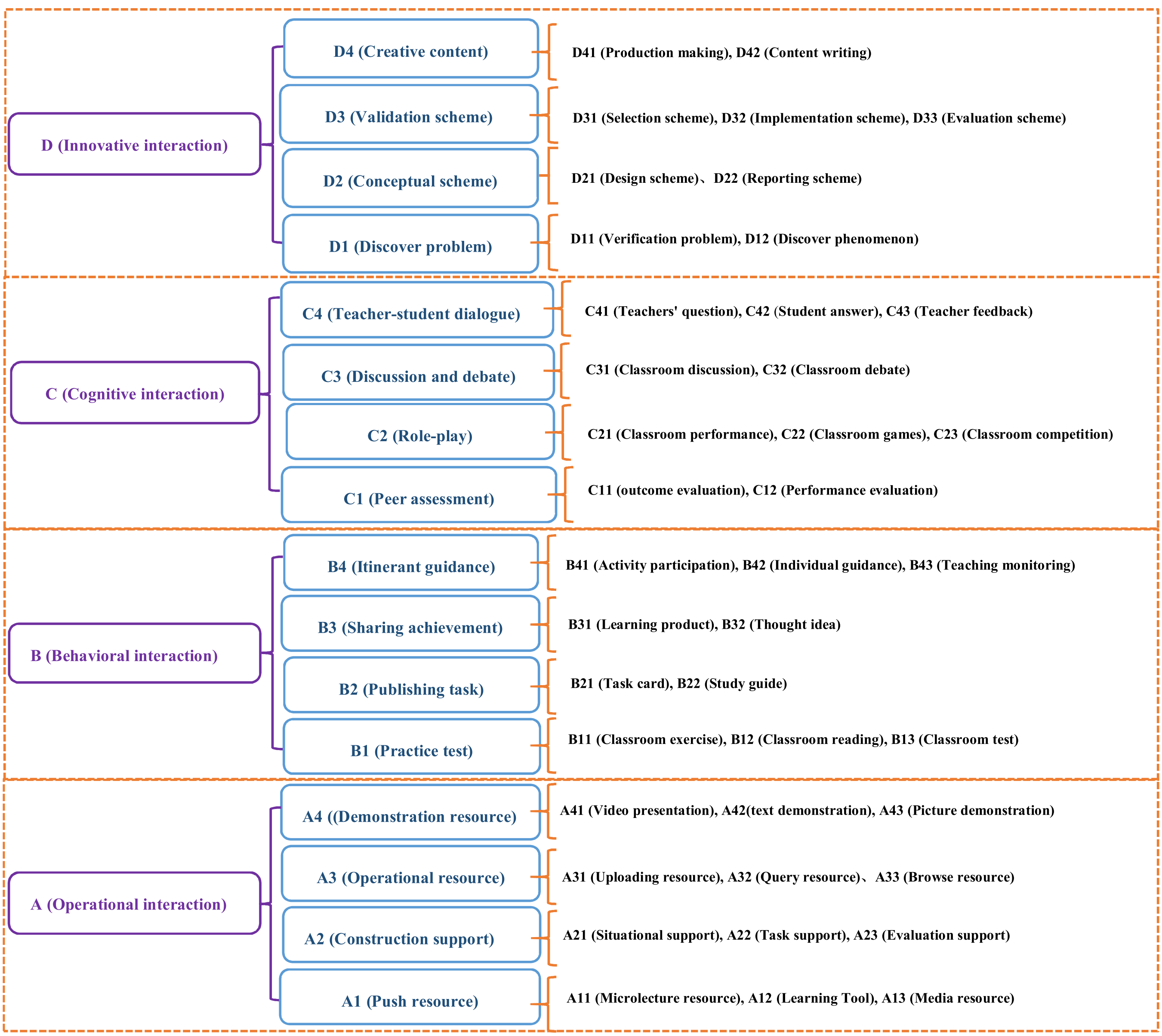

Fig. 2. Preliminary evaluation index system of TSI in SC.

3. Results and discussion

3.1. Revision of evaluation index system

In order to further improve the initially constructed evaluation system, this study used the generally recognized Delphi method (Pan et al., 2021) to revise the evaluation index system through two rounds of anonymous back-to-back consultations of experts, and finally formed a relatively complete evaluation system for TSI in SCs.

3.1.1. Expert selection

The selection of experts was one of the key factors affecting its effect and quality in the Delphi method. The selected experts were mainly senior professors and doctors from universities or research institutions familiar with the field. The number of experts selected was generally about 15–50 (Li et al., 2020). In this study, 20 experts were selected for consultation based on various factors. Among them, 18 experts had doctoral degrees, which accounted for 90% of the total number. Eighteen of these experts had doctoral degrees, which accounted for 90%. There were 17 professors, which accounted for 85%. The number of people who had worked for more than 20 years was 15, which accounted for 75%. Details of experts are shown in Table 1.

3.1.2. The first round of index revision

The purpose of the first round of expert consultation was to ask for the division of evaluation indicators, the opinions expressed in language and the scoring of importance. In order to ensure the reliability of the expert consultation results, the researchers first used the expert authority coefficient Cr to analyze the reliability of the expert scoring. The value of Cr depended on the sum of expert proficiency coefficient Cs and all judgment coefficients Ca, and Cr> 0.7 had a high reliability (Hill & Fowles, 1975). The calculation of Cr (0.855) showed that the reliability of expert consultation results was high.

Experts' feedback on the evaluation system was summarized as "modify indicator name", "add indicator item" and "delete indicator item". Experts believed that "activity participation (B41)" needed to be deleted. This indicator had a lot of overlap with "individual guidance (B42)". Teachers needed to participate in students' inquiry activities to provide personalized guidance services to students. The "teachers' question (C41)" was revised to "problem design", "students' answer (C42)" was revised to "question answering", and "teachers' feedback (C43)" was revised to "problem feedback". The technical tools in the SC extended the way of teacher-student dialogue. Nonverbal forms such as problem design, question answering and question feedback replaced traditional verbal forms such as teachers' question, student answer and teacher feedback (Nai, 2022). The "learning program (B33)" was added to "sharing achievement (B3)". Design a learning program belonged to the category of learning achievements. The "browse resource (A33)" had been changed to "download resource". There were many overlapping parts between this indicator and query resource. In some interactive links, students needed to download digital learning resources. The "video demonstration (A41)", "text demonstration (A42)" and "picture demonstration (A43)" were modified to "media demonstration", "demonstration experience" and "virtual demonstration" respectively. Teachers used media resources, subject tools, virtual simulation tools and other to demonstrate contents. The "discover phenomenon (D12)" was changed to "analyze problem". The meaning of "discover phenomenon" was not clear. The "analyze problem" instead of "discover phenomenon" to make it more specific (Su, 2021). The "creative content (D4)" was changed to "creative product". The learning content created was essentially a learning product, and learning product was used instead of learning content to maintain consistency.

Table 1
Basic information of Delphi experts participating in two rounds.

Basic information		Number	Percentage
Education background	Doctor	18	90%
	Master	2	10%
Title	Professorial	17	85%
	Associate professor	3	15%
Working years	10–20 year	5	25%
	> 20 year	15	75%

3.1.3. The second round of index revision

The procedure of the second round of expert consultation was the same as that of the first round. The first round of revised evaluation indicators continued to be consulted. Only 7 experts put forward suggestions for modification in the second round, which mainly focused on indicator description and added some indicators. The "innovative interaction (D)" was changed to "creative interaction", which better summarized the indicators. The "creative microlecture (D43)" was added to the index. Microlecture was also a typical learning product. Creative microlecture cultivated students' creative thinking. The "verification scheme (D23)" had been added. The verification scheme was an important part of the conceptual scheme. Whether the verification scheme was correct needed to be demonstrated. The "ask question" had been added, which needed to be adjusted to D11. One or more questions needed to be clearly raised after analyzing the problem (Kim et al., 2015). The "retrieval resources (A3)" was modified to "operational resources", which had a broader meaning and summarized the three indicators. The "production making (D41)" and "content writing (D42)" were revised to "creative work" and "creative article", which was more accurate. The evaluation system of TSI in SC had been revised in two rounds from the original 4 first level indicators, 16 s level indicators and 41 third level indicators to 4 first level indicators, 16 s level indicators and 44 third level indicators. Each indicator item is shown in Fig. 3, and the specific description of each indicator is shown in Table S3.

Operational interaction referred to the process in which teachers and students operated and processed learning resources with the support of intelligent platforms or terminals, which included demonstration resource, operational resource, construction support and push resource (Machado et al., 2015). Demonstration resource referred to the teaching contents that teachers used demonstration tools to demonstrate with Media player, demonstration experiment, virtual experience, etc. The operational resource referred to students' use of technical tools to upload, find and download learning resources (Huang et al., 2019). The "construction support" referred to that teachers provided students with task support, evaluation support and situation support to help them complete the inquiry task. This is beneficial for helping students improve their learning efficiency (Zhang & Li, 2021). The "push resource" referred to that teachers pushed microlecture resources, various digital learning resources and tools to students. The purpose of operational interaction was to support behavioral interaction, cognitive interaction, and creative interaction, which pointed to the cognitive goal of knowing and understanding levels.

The "Behavioral interaction" referred to various forms and natures of nonverbal interaction activities between teachers and students with the support of intelligent platforms or terminals, which included practice test, publishing task, sharing achievement and itinerant guidance (Janicke et al., 2008). The "practice test" referred to that students used the exercise and answer software to do classroom exercises, test learning achievements, and follow sentences or phrases. The "publishing task" meant that teachers published learning tasks through intelligent platforms or terminals, and students carried out inquiry activities according to task requirements. The "sharing achievement" meant that students used intelligent platforms or terminals to share thought ideas, learning products, learning programs, etc. with teachers or other students. The "tour guidance" referred to the tour guidance conducted by teachers with handheld mobile intelligent terminal devices to provide timely guidance and help to students. The objective of behavioral interaction was to support cognitive interaction and creative interaction, which

Fig. 3. Final evaluation index system of TSI in SC.

pointed to the cognitive goals of knowing, understanding and application levels (Hsu, 2022).

The "cognitive interaction" referred to various forms and natures of verbal dialogue and communication activities between teachers and students by intelligent platforms or terminals, which included peer assessment, role-play, discussion and debate, and teacher-student dialogue (Andrade et al., 2022). Peer assessment meant that students used evaluation tools to evaluate their peers' learning achievements or performance. Role-play referred to that students participated in teaching activities with different identities and roles by taking intelligent platforms or terminals as carriers. Discussion and debate referred to students' communication activities such as discussion and debate based on intelligent platform or terminal in groups. The objective of cognitive interaction was to support creative interaction, which pointed towards cognitive goals at the level of analysis and evaluation (Fu, 2023).

The "creative interaction" referred to the process of TSI solving complex problems with the support of intelligent platforms or terminals, which included discovery problems, conceptual scheme, validation scheme and creative products (Mellati et al., 2018). Discover problems referred to students' use of discipline inquiry tools to analyze problems, put forward problems and demonstrate problems. Conceptual scheme referred to the process that students used discipline inquiry tools to conceive multiple solutions, and design, report and demonstrate them to solve problems. Validation scheme referred to the process that students used discipline inquiry tools to demonstrate and evaluate the proposed solutions. Creative product referred to students' use of creative work, creative article, and creative microlecture. Creative interaction was the highest goal of TSI in the SC, which pointed to the cognitive goal of analysis and evaluation, especially at the creative level (Corella, 2020).

3.2. *Weight calculation of evaluation index system and the advantages analysis of the evaluation index system*

Two rounds of expert opinions were solicited, the evaluation system of TSI in SC was composed of 4 first level indicators, 16 s level indicators, 44 third level indicators and specific instructions. The index weight was calculated to increase the operability of the evaluation system. The questionnaire was designed by using the AHP (Naveed et al., 2023), and the index weight was judged by using the method of group decision-making to avoid the limitation and one-sided of the index weight. The 20 experts in the first and second rounds participated in the AHP decision-making, and a total of 15 valid questionnaires were recovered with a recovery rate of 75%.

3.2.1. Calculation of index weight value

According to the recovered questionnaire, the relative importance of the primary indicators was calculated. In order to effectively calculate the relative weight value of indicators at all levels, the AHP was used to assist the software Yaahp for analysis. The final weight value of each evaluation index was obtained through consistency test. Table 2 was the judgment matrix of the relative importance of one of the experts. The analysis of the judgment matrix determined the consistency of expert scoring, and the assignment of the judgment matrix was normalized twice to finally obtain the initial weight value of the primary index.

3.2.2. Index consistency test and determination of index weight

This study used the CR value of consistency to test whether the scoring results were consistent. The consistency of each expert's score was lower than 0.1, which indicated that the expert's judgment results were internally consistent and scientific (Zhai & Dong, 2022). Yaahp AHP software was used for processing and analysis. When the consistency of the 15 expert judgment matrices was analyzed, the weights of the 15 judgment matrices were calculated respectively, and then the weighted mean method was used to process the results to finally obtain the weight values of each indicator. The weight values of the first level indicators are shown in Table 3.

According to the above steps, this study obtained the single level weight coefficient values of the first, second and third level indicators of TSI evaluation system in SC (Table 4).

3.2.3. Analysis of the evaluation indicator system advantages

Compared with other studies, the evaluation system for TSI in SC constructed in this article had both consensus and differences. Some scholars divided the evaluation indicators of TSI in SC into two primary dimensions (teacher-student language interaction and teacher-student technical interaction), and each primary dimension included two secondary dimensions (Zhang et al., 2016). Previous studies have mainly focused on teacher-student behavioral interaction and language behavioral interaction, less attention was paid to cognitive and creative interactions that were conducive to the development of students' HOT, and each indicator lacked effective support from intelligent technology (Corella, 2020). This study comprehensively constructed an evaluation index system that was conducive to the development of students' HOT from four dimensions (operational interaction, behavioral interaction, cognitive interaction, and creative interaction). Moreover, each index requires effective support from intelligent technology. Therefore, this study was an extension of the existing research, which was significantly higher than existing research. The evaluation indicators for TSI in SC constructed in this study fully reflect the characteristics of SC (Cebrián et al., 2020).

3.3. Application of evaluation index system

In order to better test the effectiveness and feasibility of the evaluation system, this study prepared the "questionnaire on the TSI in SC". The 214 primary and secondary school teachers from 9 provinces in eastern, central and western China were selected for the survey. The respondents involved many subjects such as physics, chemistry, mathematics, Chinese, geography, and morality, which included three learning stages of primary school, junior high school and senior high school. The 214 questionnaires were distributed through the questionnaire star platform, 197 questionnaires were collected within the specified time, and 189 valid questionnaires were collected. These questionnaires were imported into SPSS 24 software for analysis. The coefficient of questionnaire overall Cronbach's α was 0.969, and the Cronbach's α of the four indicators of operational interaction, behavioral interaction, cognitive interaction, and creative interaction were 0.903, 0.888, 0.852 and 0.840 respectively. The Cronbach's α was greater than 0.7, the indicator had good reliability (West et al., 2010), and the questionnaire designed in this study had good reliability. KMO and Bartlett's spherical test (Nahandi et al., 2020) were 0.858 ($P < 0.001$), The 44 items contained in the questionnaire were well aggregated into 4 common factors, which indicated that the designed questionnaire had good structural validity.

The specific operation was to divide the evaluation indicators into five grades: A, B, C, D and E by using a rating scale (Mergany et al., 2021). The values were assigned in turn as 5, 4, 3, 2 and 1 for the convenience of statistical analysis. A, B, C, D, and E indicated complete conformity, conformity, relatively conformity, less conformity, complete non-conformity respectively. The teacher only needed to give a score according to the degree of compliance with the indicators, which converted the score into a composite score according to the teacher's rating value to achieve two quantifications. The mean score of all teachers was calculated by the evaluation score of the three level indicators. The evaluation score of the second level index was the sum of the evaluation score of the third level index multiplied by the weight coefficient of the third level index. The evaluation score of primary indicators was the sum of the evaluation score of secondary indicators multiplied by the weight coefficient of secondary indicators. The horizontal values of each dimension indicator of operational interaction, behavioral interaction, collaborative interaction, and creative interaction were calculated (Fig. 4).

The horizontal values of "operational interaction (A)" and "behavioral interaction (B)" were on the high side, while the horizontal values of "C (cognitive interaction)" and "D (creative interaction)" were on the low side. This showed that Chinese primary and secondary school teachers and students often used intelligent devices to support activities such as push resource, construction support, demonstration resource, itinerant guidance, sharing result and publishing task, and rarely used intelligent devices to support peer evaluation, teacher-student dialogue, discussion and debate, role-play, discover problem, conceptual scheme, validation scheme, and creative product. Operational interaction and behavioral interaction mainly cultivated students to know, understand and apply lower order thinking (Jou et al., 2016), while cognitive interaction and creative interaction mainly cultivated students to analyze, evaluate and create HOT (Fu, 2023). The level of TSI in primary and secondary schools in China, especially the level of collaborative interaction and creative interaction, needed to be further improved.

The three indicators "task support (A22)", "download resource (A33)" and "virtual demonstration (A43)" included in the first level indicator "operational interaction (A)" scored the lowest, which indicated that teachers rarely provided students with task supports, required students to download learning resources and used virtual simulation tools to demonstrate teaching content. The two indicators "D13 (analyze problem)" and "D43 (creative microlecture)" included in the first level indicator "D (creative interaction)" scored the lowest, which indicated that teachers rarely required students to analyze problems and create microlecture. This indicates that the teacher's requirements for students

Table 2
First level index judgment matrix (expert 1).

Level I indicator	A (Operational interaction)	B (Behavioral interaction)	C (Cognitive interaction)	D (Creative interaction)
A (Operational interaction)	1	1	1/3	1/5
B (Behavioral interaction)	1	1	1	1/3
C (Cognitive interaction)	3	1	1	1
D (Creative interaction)	5	3	1	1

Table 3
Weighted value of primary indicators.

Expert code	A (Operational interaction)	B (Behavioral interaction)	C (Cognitive interaction)	D (Creative interaction)	CR
Z1	0.112	0.167	0.289	0.432	0.076
Z2	0.053	0.104	0.253	0.590	0.052
Z3	0.049	0.099	0.312	0.540	0.085
Z4	0.319	0.322	0.181	0.179	0.096
Z5	0.117	0.068	0.285	0.530	0.097
Z6	0.056	0.101	0.215	0.621	0.095
Z7	0.145	0.230	0.309	0.320	0.094
Z8	0.191	0.423	0.309	0.077	0.100
Z9	0.250	0.155	0.536	0.058	0.097
Z10	0.168	0.188	0.464	0.181	0.098
Z11	0.092	0.092	0.203	0.606	0.057
Z12	0.117	0.068	0.285	0.530	0.097
Z13	0.064	0.150	0.312	0.674	0.098
Z14	0.340	0.186	0.241	0.233	0.097
Z15	0.114	0.222	0.140	0.524	0.087

Table 4
Weight coefficient of evaluation index for TSI in SC.

Level I index	Weight coefficient	Level 2 index	Weight coefficient	Level 3 index	Weight coefficient
A (Operational interaction)	0.1329	A1 (Push resource)	0.1947	A11 (Microlecture resource)	0.3529
				A12 (Learning Tool)	0.2942
				A13 (Media resource)	0.3529
		A2 (Construction support)	0.2253	A21 (Situational support)	0.3037
				A22 (Task support)	0.3530
				A23 (Evaluation support)	0.3433
		A3 (Operational resource)	0.2948	A31 (Uploading resource)	0.3136
				A32 (Query resource)	0.3136
				A33 (Download resource)	0.3718
		A4 ((Demonstration resource)	0.2852	A41 (Media player)	0.3138
				A42 (Demonstration experiment)	0.2845
				A43 (Virtual demonstration)	0.4017
B (Behavioral interaction)	0.1731	B1 (Practice test)	0.2451	B11(Classroom exercise)	0.3330
				B12 (Classroom reading)	0.3140
				B13 (Classroom test)	0.2940
		B2 (Publishing task)	0.2349	B21 (Task card)	0.4900
				B22 (Study guide)	0.5100
		B3 (Sharing achievement)	0.2430	B31 (Learning product)	0.3140
				B32 (Thought idea)	0.3045
				B33 (Learning programme)	0.3815
		B4 (Itinerant guidance)	0.2770	B41 (Individual guidance)	0.5300
				B42 (Teaching monitoring)	0.4700
C (Cognitive interaction)	0.2978	C1 (Peer assessment)	0.2146	C11 (Outcome evaluation)	0.490
				C12 (Performance evaluation)	0.510
		C2 (Role-play)	0.2500	C21 (Classroom performance)	0.3533
				C22 (Classroom games)	0.3034
				C23 (Classroom competition)	0.3433
		C3 (Discussion and debate)	0.2500	C31 (Classroom discussion)	0.4710
				C32 (Classroom debate)	0.5290
		C4 (Teacher-student dialogue)	0.2854	C41 (Problem design)	0.3237
				C42 (Problem answering)	0.3237
				C43 (Problem feedback)	0.3536
D (Creative interaction)	0.3962	D1 (Discover problem)	0.2448	D11 (Ask question)	0.3140
				D12 (Verification problem)	0.3140
				D13 (Analyze problem)	0.3330
		D2 (Conceptual scheme)	0.2600	D21 (Design scheme)	0.3243
				D22 (Reporting scheme)	0.3142
				D23 (Verification scheme)	0.3625
		D3 (Validation scheme)	0.2400	D31 (Selection scheme)	0.3140
				D32 (Implementation scheme)	0.3240
				D33 (Evaluation scheme)	0.3630
		D4 (Creative product)	0.2552	D41 (Creative work)	0.4220
				D42 (Creative article)	0.3140
				D43 (Creative microlecture)	0.2650

are not strict enough (Zhan et al., 2021b).

The scores of "B33 (learning program)" and "B41 (individual guidance)" included in the first level indicator "B (behavioral interaction)" were not high, which indicated that teachers did not pay enough attention to students' requirements for sharing learning programs and providing personalized guidance to students. The "C (cognitive interaction)" included "C11 (outcome evaluation)", "C21 (classroom performance)", "C23 (classroom competition)", "C31 (classroom discussion)", and "C43 (problem feedback)" with low scores, which indicated that teachers were insufficient in requiring students to evaluate peer achievements, classroom performance, classroom debate and classroom competitions. Teachers rarely gave timely feedback and

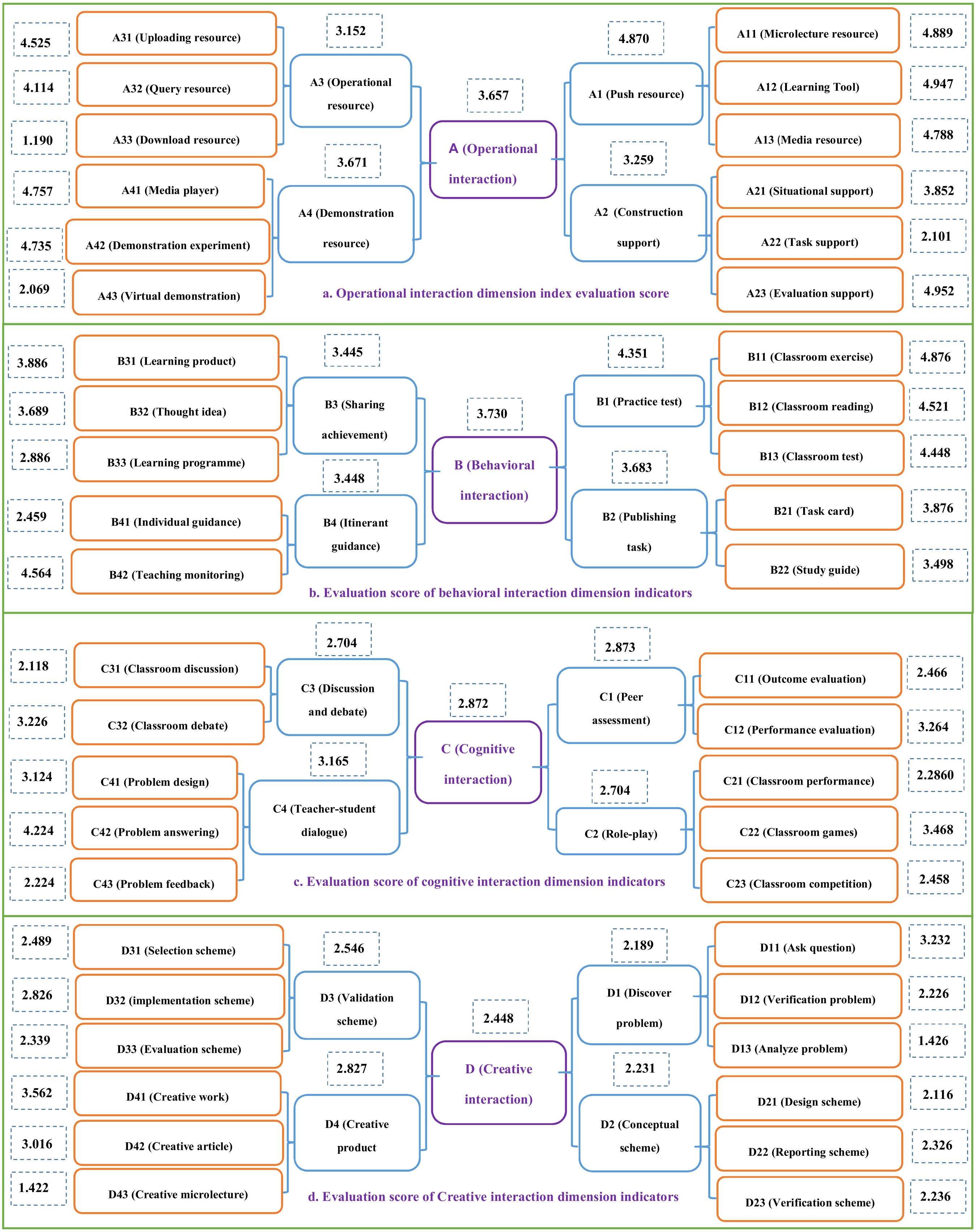

Fig. 4. Evaluation scores of each dimension index of TSI in SC.

evaluation on students' answers to questions (Hsu, 2022). The scores of seven indicators "verification problem (D12)", "design scheme (D21)", "reporting scheme (D22)", "verification scheme (D23)", "selection scheme (D31)", "implementation scheme (D32)" and "evaluation scheme (D33)" included in "creative interaction (D)" were also not high. This showed that teachers rarely required students to use discipline tools to demonstrate problems, design scheme, report scheme, verify scheme, select scheme, implement scheme, and evaluate scheme (Lee, 2011).

While applying the indicator system, the researchers also received feedback from teachers. Most teachers thought that the design of the indicator system was more reasonable, and the distribution of indicator scores was more scientific, which reflected the actual demand for TSI in SC in primary and secondary schools. The evaluation index system and scoring table were easy to use and highly operable (Dai et al., 2021). In general, the survey results were highly consistent with the teachers' subjective feelings, which confirmed that there were some problems of TSI in the SC for primary and secondary schools that needed to be further optimized and adjusted.

4. Conclusions and outlook

The grounded theory method was used to code and analyze the teaching video of SC, initially construct the evaluation system of TSI in SC. Delphi method and AHP were used to optimize the scientific and rationality of the evaluation indicators, which finally formed a scientific evaluation system for TSI in SC. The evaluation system included four first level indicators (operational interaction, behavioral interaction, cognitive interaction, and creative interaction), 16 s level indicators, 44 third level indicators and indicator descriptions.

From the perspective of the weight of the first level indicators, the weight assignment reflected the importance of the indicators, which was more consistent with the theoretical significance and practical operation. In each second level and third level indicator, the collaborative interaction and creative interaction were significantly higher than the operational interaction and behavioral interaction. The cognitive interaction and creative interaction were the key element of the TSI in the SC and the most concerned content in the evaluation of TSI in the SC, which were also the central link of TSI and communication activities.

Overall, the evaluation system comprehensively evaluated each element of TSI in the SC and made up for the lack of research on TSI in the SC. This evaluation system provided theoretical guidance for teachers to build high-quality and effective SC and had important theoretical value and practical significance for the development of students' HOT and the cultivation of innovative talents.

There were still deficiencies in this study. Although the selection of evaluation indicators for TSI in SC has been carefully selected by grounded theory, there has not been much consensus in the field of TSI in SC. The initial indicators selected may limit the proposal of expert modification opinions, so it needs further testing. From the perspective of evaluation tools, this study adopted a self-evaluation form in terms of application, which has limitations in evaluating the actual level of TSI. Realistic and objective evaluation of the effectiveness and quality of TSI still relies on real classroom situations, and self-evaluation tools can serve as one of the multimodal proofs of triangular mutual verification. In addition, this research did not consider specific disciplines and learning stages from the perspective of factor view. Subsequent research will need to build a more targeted evaluation index system for specific disciplines and learning stages to better guide classroom teaching practice.

CRediT authorship contribution statement

Xundiao Ma drafted the initial manuscript, performed the experiments, analyzed data. Yueguang Xie carried out the initial analyses, reviewed and revised the manuscript. Hanxi Wang conceptualized, designed, reviewed, and revised the manuscript. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.stueduc.2023.101286.

References

Alfadil, M., Anderson, D., & Green, A. (2020). Connecting to the digital age: using emergent technology to enhance student learning. *Education and Information Technologies, 25*, 1625–1638. https://doi.org/10.1007/s10639-019-10035-z

Andrade, V. D., Freire, S., Baptista, M., & Shwartz, Y. (2022). Drawing as a space for social-cognitive interaction. *Education Science, 12*, 45. https://doi.org/10.3390/educsci12010045

Balmer, D. F., & Richards, B. F. (2017). Longitudinal qualitative research in medical education. *Perspectives on Medical Education, 6*(5), 306–310. https://doi.org/10.1007/s40037-017-0374-9

Buell, M., Han, M., & Vukelich, C. (2017). Factors affecting variance in Classroom Assessment Scoring System scores: season, context, and classroom composition. *Early Child Development and Care, 187*(11), 1635–1648. https://doi.org/10.1080/03004430.2016.1178245

Cebrián, G., Palau, R., & Mogas, J. (2020). The Smart classroom as a means to the development of ESD methodologies. *Sustainability, 12*, 3010. https://doi.org/10.3390/su12073010

Chen, H., Nunes, M. B., Ragsdell, G., & An, X. (2018). Extrinsic and intrinsic motivation for experience grounded tacit know ledge sharing in Chinese software organisations. *Journal of Knowledge Management, 22*(2), 478–498. https://doi.org/10.1108/JKM-03-2017-0101

Corella, M. (2020). Talking "smart": Academic language and indexical competence in peer interactions in an elementary classroom. *Linguistics and Education, 55*, Article 100755. https://doi.org/10.1016/j.linged.2019.100755

Dai, Z., Sun, C., Zhao, L., & Li, Z. (2021). Assessment of smart learning environments in higher educational institutions: A study using AHP-FCE and GA-BP methods. *IEEE Access, 9*, 35487–35500. https://doi.org/10.1109/ACCESS.2021.3062680

Flanders, N. A. (1963). Intent, action and feedback: A preparation for teaching. *Journal of Teacher Education, 14*(3), 251–259. https://doi.org/10.1177/002248716301400305

Fu, S. (2023). A reinforcement learning-based smart educational environment for higher education. *International Journal of e-Collaboration, 19*(6), 1–17. https://doi.org/10.4018/IJeC.315019

He, Y., Lu, J., Huang, H., He, S., Ma, N., Sha, Z., Sun, Y., & Li, X. (2019). The effects of flipped classrooms on undergraduate pharmaceutical marketing learning: A clustered randomized controlled study. *PLoS ONE, 14*(4), Article e0214624. https://doi.org/10.1371/journal.pone.0214624

Hill, K. Q., & Fowles, J. (1975). The methodological worth of the Delphi forecasting technique. *Technological Forecasting and Social Change, 7*(2), 179–192. https://doi.org/10.1016/0040-1625(75)90057-8

Hsu, T. C. (2022). The different effects of daily-life instant response social media and an educational feedback system on flipped learning: from the evidence of behavioral analysis. *Interactive Learning Environments, 30*(5), 862–881. https://doi.org/10.1080/10494820.2019.1692358

Huang, L., Su, J., & Pao, T. L. (2019). A context aware smart classroom architecture for smart campuses. *Applied Sciences, 9*, 1837. https://doi.org/10.3390/app9091837

Huang, W., & Tang, C. (2020). Innovative application of information-based education in primary and secondary schools in the post-epidemic era: Beginner's mind, philosophy and path. *China Educational Technology, 41*(12), 124–130. https://doi.org/10.3969/j.issn.1006-9860.2020.12.018

Janicke, D. M., Mitchell, M. J., Quittner, A. L., Piazza-Waggoner, C., & Stark, L. J. (2008). The impact of behavioral intervention on family interactions at mealtime in pediatric cystic fibrosis. *Children's Health Care, 37*(1), 49–66. https://doi.org/10.1080/02739610701766891

Jou, M., Tennyson, R. D., Wang, J., & Huang, S. Y. (2016). A study on the usability of E-books and APP in engineering courses: A case study on mechanical drawing. *Computers & Education, 92–93*, 181–193. https://doi.org/10.1016/j.compedu.2015.10.004

Kim, C., Kim, J., Park, H., & Ha, I. (2015). Analysis of college classes based on U-CLASS system using personal mobile nodes. *International Journal of Distributed Sensor Networks, 11*(5), Article 493407. https://doi.org/10.1155/2015/493407

Lee, I. (2011). Working smarter, not working harder: Revisiting teacher feedback in the L2 writing classroom. *Canadian Modern Language Review, 67*(3), 377–399. https://doi.org/10.3138/cmlr.67.3.377

Li, Y., Dong, X., Li, X., & Zhang, Y. (2020). Construction of quality evaluation indicator system for STEM education——Based on the Delphi method and analytic hierarchy process. *Modern Distance Education Research, 32*(2), 48–55+72. https://doi.org/10.3969/j.issn.1009-5195.2020.02.006

Li, Feng, & Wu, Z. (2022). Research on the composition of national security discipline core literacy based on grounded theory: Taking the United States homeland security discipline as reference. *Journal of Intelligence, 41*(6), 80–85. https://doi.org/10.3969/j.issn.1002-1965.2022.06.012

Machado, R., Ott, P. K., Moreno, I. B., Danilewicz, D., Tavares, M., Crespo, E. A., Siciliano, S., & De Oliveira, L. R. (2015). Operational interactions between South American sea lions and gillnet fishing in southern Brazil. *Aquatic Conservation: Marine Freshwater Ecosystems, 26*(1), 108–120. https://doi.org/10.1002/aqc.2554

Marian, H., & Jackson, C. (2017). Inquiry-based learning: a framework for assessing science in the early years. *Early Child Development and Care, 187*(2), 221–232. https://doi.org/10.1080/03004430.2016.1237563

Mellati, M., Khademi, M., & Abolhassani, M. (2018). Creative interaction in social networks: Multi-synchronous language learning environments. *Education and Information Technologies, 23*(5), 2053–2071. https://doi.org/10.1007/s10639-018-9703-9

Mergany, N. N., Dafalla, A. E., & Awooda, E. (2021). Effect of mobile learning on academic achievement and attitude of Sudanese dental students: a preliminary study. *BMC Medical Education, 21*, 121. https://doi.org/10.1186/s12909-021-02509-x

Nai, R. (2022). The design of smart classroom for modern college English teaching under Internet of Things. *PLoS ONE, 17*(2), Article e0264176. https://doi.org/10.1371/journal.pone.0264176

Naveed, Q. N., Qahmash, A. I., Qureshi, M. R. N., Ahmad, N., Abdul Rasheed, M. A., & Akhtaruzzaman, M. (2023). Analyzing critical success factors for sustainable cloud-based mobile learning (CBML) in crisp and fuzzy environment. *Sustainability, 15*, 1017. https://doi.org/10.3390/su15021017

Nahandi, M. Z., Jafarabadi, M. A., Haghighatjou, M., Ashrafrezaei, K., Saeedabad, M. S., Mohammadi, A. B., & Mohammadi, A. B. (2020). Attitudes of patients and their families towards medical privacy and competence of bearer or receiver of bad news. *International Journal of Medical Toxicology and Forensic Medicine, 10*(4), 28580. https://doi.org/10.32598/ijmtfm.v10i4.28580

Pan, L., Tlili, A., Li, J., Jiang, F., Shi, G., Yu, H., & Yang, J. (2021). How to implement game-based learning in a smart classroom? A model based on a systematic literature review and Delphi method. *Frontiers in Psychology, 12*, Article 749837. https://doi.org/10.3389/fpsyg.2021.749837

Sawatsky, A. P., Ratelle, J. T., & Beckman, T. J. (2019). Qualitative research methods in medical education. *Anesthesiology, 131*(1), 14–22. https://doi.org/10.1097/ALN.0000000000002728

Shi, Y., Peng, C., Zhang, J., & Yang, H. (2019). Research on the teacher-student interaction behavior in colleges and universities under smart classroom environment. *Modern Educational Technology, 29*(1), 45–51. https://doi.org/10.3969/j.issn.1009-8097.2019.01.007

Su, H. (2021). A research-based smart classroom in an exercise physiology course. *International Journal of Emerging Technologies in Learning, 16*(18), 97–112. https://doi.org/10.3991/ijet.v16i18.25339

Wang, Wan, Zhang, X., Guo, S., & Wu, Y. (2022). Construction of user demand model of mobile professional virtual community based on grounded theory. *Information Science, 40*(6), 169–176. https://doi.org/10.13833/j.issn.1007-7634.2022.06.021

West, S. G., Thoemmes, F., & Wu, W. (2010). Introduction to structural equation modelling: Using SPSS and AMOS. *Applied Psychological Measurement, 34*(3), 211–213. https://doi.org/10.1177/0146621609360954

Wu, Y., Wu, G., & Ma, S. (2016). Review on the origin, genre and application of the grounded theory method. *Journal of Distance Education, 35*(3), 32–41. https://doi.org/10.15881/j.cnki.cn33-1304/g4.2016.03.004

Xu, L., & Yang, Q. (2019). Modeling and analysis on teacher-student relationship. *Discrete Dynamics in Nature and Society, 2019*, Article 5481926. https://doi.org/10.1155/2019/5481926

Xue, E., Li, J., & Xu, L. (2022). Online education action for defeating COVID-19 in China: An analysis of the system, mechanism and mode. *Educational Philosophy and Theory, 54*(6), 799–811. https://doi.org/10.1080/00131857.2020.182118

Yuan, Y. (2022). Quantitative analysis of Chinese classroom teaching activity under the background of artificial intelligence. *Education Information Technology, 27*, 11161–11177. https://doi.org/10.1007/s10639-022-11080-x

Zhai, Z., & Dong, B. (2022). The construction of evaluation index system for secondary school Chinese textbooks: A study based on Delphi method. Curriculum. *Teaching Material and Method, 42*(9), 129–135. https://doi.org/10.19877/j.cnki.kcjcjf.2022.09.005

Zhan, Z., Wu, Q., Lin, Z., & Cai, J. (2021a). Smart classroom environments affect teacher-student interaction: Evidence from a behavioural sequence analysis. *Australasian Journal of Educational Technology, 37*(2), 96–109. https://doi.org/10.14742/ajet.6523

Zhan, Z., Wu, Q., He, W., Cheng, S., Lu, J., & Han, Y. (2021b). K12 teacher-student interaction patterns in the smart classrooms. *International Journal of Innovation and Learning, 29*(3), 267–286. https://doi.org/10.1504/IJIL.2021.114511

Zhang, Y., Zhu, Y., Bai, Q., Zhu, Y., & Li, X. (2016). Research of the teaching interaction behavior characteristics of primary mathematics in the smart classroom. *China Educational Technology, 37*(6), 43–48+64. https://doi.org/10.3969/j.issn.1006-9860.2016.06.009

Zhang, B., & Zhang, Q. (2018). Research on influencing factors of users' behavior intention of knowledge contribution in social Q & A community based on grounded theory. *Journal of the China Society for Scientific and Technical Information, 37*(10), 1034–1045. https://doi.org/10.3772/j.issn.1000-0135.2018.10.007

Zhang, M., & Li, X. (2021). Design of smart classroom system based on internet of things technology and smart classroom. *Mobile Information Systems, 2021*, Article 5438878. https://doi.org/10.1155/2021/5438878

Zhang, R., & Tian, L. (2022). Teacher design thinking in smart classroom: Structural elements and cultivation paths. e-Education. *Research, 43*(9), 106–113. https://doi.org/10.13811/j.cnki.eer.2022.09.013

Zong, M., Cui, Y., & Tian, J. (2020). Research on the practice effect of "Suspending classes without stopping learning" in experimental schools of digital campuses——A study based on the investigation and analysis of 107 schools in Hubei Province. *Modern Educational Technology, 30*(8), 106–112. https://doi.org/10.3969/j.issn.1009-8097.2020.08.014